



S Lab Sync — Setup Guide

Everything you need to install on a fresh computer before running the app

Before You Start: The Checklist

S Lab Sync is a Java desktop application that stores its data in a MySQL database. Neither Java nor MySQL comes bundled with the app, so on a brand-new computer, three things need to be installed or in place before the app will run:

#	Requirement	Why it's needed
1	Java 17 or newer (JRE/JDK)	The app itself is a Java program — without Java installed, the .jar file simply won't launch.
2	MySQL Server	All student, teacher, equipment, chemical and experiment data is stored in a MySQL database, so a MySQL server has to be running somewhere the app can reach.
3	MySQL Connector/J (JDBC driver)	Java cannot talk to MySQL on its own — this small driver file bridges the two, and has to be placed in the app's lib folder.

Each of these is a one-time setup per computer. Once installed, launching the app afterwards is just a double-click.

STEP 1 Install Java (17 or newer)

Check first whether Java is already installed. Open a terminal / command prompt and run:

```
java -version
```

If this prints a version number of 17 or higher, Java is already set up — skip to Step 2. Otherwise, install it:

- Go to <https://adoptium.net> (Eclipse Temurin — a free, official OpenJDK build).
- Download the installer for your operating system (Windows, macOS, or Linux) — choose the **JRE** if you only want to run the app, or the **JDK** if you also plan to modify/recompile the source code.
- Run the installer, accepting the default options.
- Close and reopen your terminal, then re-run `java -version` to confirm it worked.

STEP 2 Install MySQL Server

The app needs a MySQL server it can connect to — this is normally installed on the same computer (`localhost`), but it can also point at a MySQL server on another machine on the network.

Windows

- Download the MySQL Installer from <https://dev.mysql.com/downloads/installer/>.
- Run it and choose the **"Server only"** (or "Developer Default") setup type.
- During setup you'll be asked to create a **root password** — remember this, it's the password you'll type into the app later.
- Let the installer configure MySQL to start automatically as a Windows service.

macOS

- Easiest via Homebrew: `brew install mysql`, then start it with `brew services start mysql`.
- Or download the official .dmg installer from the same MySQL downloads page and follow its prompts.
- Set a root password with `mysql_secure_installation` if the installer doesn't prompt for one.

Linux (Debian/Ubuntu-based)

```
sudo apt update
sudo apt install mysql-server
sudo systemctl enable --now mysql
sudo mysql_secure_installation
```

Whichever platform, make a note of the **username** and **password** you set — you'll need an account that has permission to create databases and tables (the default root account works fine).

STEP 3 Add the MySQL Connector/J driver

Java doesn't know how to talk to MySQL out of the box — it needs a small JDBC driver file. This is a one-time download.

- Go to <https://dev.mysql.com/downloads/connector/j/>.
- Under "Select Operating System", choose "**Platform Independent**".
- Download the .zip (or .tar.gz) archive.
- Extract it, and find the file named something like `mysql-connector-j-9.x.x.jar`.
- Copy that file into the S Lab Sync `lib` folder and **rename it** to exactly `mysql-connector-j.jar` — the launcher looks for this exact name.

The folder should end up looking like this:

```
SLabSync.jar
Run_SLabSync.bat
Run_SLabSync.sh
logo.ico
lib/
  mysql-connector-j.jar <-- the file you just added
```

If this file is missing or misnamed, the app will start but fail with a database connection error the moment it tries to connect — this is the most common first-run problem.

STEP 4 Run S Lab Sync for the first time

- **Windows:** double-click `Run_SLabSync.bat`.
- **macOS / Linux:** open a terminal in the app folder and run `./Run_SLabSync.sh` (or double-click it, if your file manager allows running scripts).

What to expect on first launch:

- A "**Connect to MySQL**" window appears first — enter the MySQL username/password from Step 2 (host defaults to `localhost`). This is saved for next time, so you only do this once per computer.
- The app automatically creates the `lab_management` database and all required tables — no manual SQL needed.
- The app's own login screen appears — sign in with username `Master` and password `7777`.
- You'll be asked whether to add a desktop shortcut (Windows only) — choose `Yes` for a one-click launch next time.

Troubleshooting Common First-Run Issues

Symptom	Likely cause & fix
App won't start at all / nothing happens when double-clicking	Java isn't installed, or isn't on the system PATH. Re-check Step 1 and confirm "java -version" works from a terminal.
Error mentioning "driver" or "ClassNotFoundException" / "No suitable driver"	The mysql-connector-j.jar file is missing from the lib folder, or was named incorrectly. Re-check Step 3 — the filename must match exactly.
"Access denied for user" when connecting	The MySQL username or password entered doesn't match what was set up in Step 2. Use "Change MySQL Login" in the sidebar to re-enter the correct details.
"Could not connect" / "Communications link failure"	MySQL Server isn't running. On Windows, check the MySQL service is started; on macOS/Linux, run "brew services start mysql" or "sudo systemctl start mysql".
Forgot the MySQL details that were saved	Open the app, click "Change MySQL Login" near the bottom of the sidebar to update or clear the saved credentials and re-enter them.

Quick Reference

Item	Where to get it
Java 17+	https://adoptium.net
MySQL Server	https://dev.mysql.com/downloads/installer/ (or your OS package manager)
MySQL Connector/J	https://dev.mysql.com/downloads/connector/j/
App login	Username: Master Password: 7777

Once Java, MySQL, and the connector jar are in place, this setup only needs to be done once per computer — every launch after that is just opening the app.